Notes: Aronoff & Rafey (AER, 2026)

Conservation Priorities and Environmental Offsets: Markets for Florida Wetlands

Contents

1 Big picture

- **Question.** How valuable are environmental offset markets, and how should they be redesigned when offsets create unpriced local externalities?
- **Setting.** The paper studies markets for Florida wetland mitigation offsets. Developers can build on protected wetlands if they buy certified offsets from wetland mitigation banks that restore or create wetlands elsewhere in the same hydrological service area.
- **Core trade-off.** Offset trading creates private gains from trade by moving conservation from high-development-value wetlands to lower-cost restoration sites. But it can also reallocate wetlands away from places where local flood protection is valuable.
- **Data contribution.** The authors assemble new data on Florida wetland development, wetland mitigation banks, offset ledgers, transaction prices, bank production schedules, contracts, flood insurance claims, flood risk, land ownership, and land use.
- **Main descriptive facts.** Wetland development tends to occur in developed, high-flood-risk areas. Wetland banks tend to locate in peripheral watersheds with more public wetlands and lower development density.
- **Market facts.** Offset sales totaled more than $1.1 billion from 1995 to 2018. Markets are concentrated: an average market has fewer than three active banks.
- **Structural approach.** The paper estimates demand for offsets, dynamic costs of wetland bank entry and production, and local flood externalities from wetland loss/restoration.
- **Main welfare result.** Relative to direct conservation, offset markets generate about $2.4 billion in private gains from trade from 1995 to 2020, but also about $1.9 billion in additional flood damages.
- **Policy result.** A locally differentiated Pigouvian tax equal to expected marginal flood damages would prevent about $1.6 billion of new flood damage while preserving more than two-thirds of the private gains from trade.
- **Economic takeaway.** Environmental offset markets can create large gains, but the correct design requires pricing location-specific externalities rather than treating all offsets as perfect substitutes.

2 Data and measurement

2.1 Wetlands and offsets

- Florida law follows a “No Net Loss” principle: development on wetlands is allowed only if wetland loss is offset by certified restoration or creation elsewhere.
- Wetland mitigation banks commit land to permanent conservation easements and restore wetlands over time.
- Regulators release offset credits gradually as restoration milestones are verified.
- Developers buy credits from banks within the same hydrological service area.

Interpretation: The market substitutes spatial reallocation for direct on-site conservation. This generates gains when restoration is cheaper elsewhere, but it can also move wetland services away from the communities exposed to development.

2.2 Geographic and market data

- The paper maps wetlands, development, public lands, wetland bank parcels, and market boundaries across Florida.
- Local units are watersheds, and markets are hydrological service areas.
- The paper observes the location and timing of wetland development from land-use transitions.
- Offset market data include bank entry, bank production schedules, offset ledgers, prices, and quantities.

Interpretation: The geography matters because wetland services are local. A credit that satisfies the regulator's aggregate conservation objective may not replace the local flood protection lost at the development site.

2.3 Flood data

- Flood outcomes are measured using National Flood Insurance Program claims.
- The paper focuses on claims after the market, especially 2016–2020, for properties built before 1995.
- This helps isolate spillovers from wetland reallocation rather than damages to newly built properties.
- Controls include baseline flood claims, flood-zone classifications, baseline development density, and nonwetland development.

Interpretation: The flood estimates use realized claims to infer how wetland loss and restoration change expected local flood damages.

2.4 Observed trade and production measures

- The median bank produces about 200 credits over its lifetime; the mean is about 410 credits.
- The median developer purchase is about 1.1 credits; the mean is about 4.1 credits.
- Credits are released slowly, with a typical release schedule implying roughly 18 years to build the full project.
- Average observed prices are roughly $88,000 per credit, while observed costs are substantially lower on average.

Interpretation: Large scale differences between banks and developers imply that supply is concentrated and dynamic. Banks are long-lived producers with inventory decisions, not passive sellers.

3 Models and empirical specifications

3.1 Regulatory conservation objective

Let $w_{it} = 1$ if location i contains a wetland at time t and let $\tilde{v}_i$ be the regulator's scalar wetland value. Aggregate regulatory wetland value is

$$\tilde{v}(w_t) = \int_0^1 w_{it}\tilde{v}_i \, di. \tag{1}$$

The No Net Loss constraint is

$$\tilde{v}(w_t) \geq \tilde{v}(w_0) \qquad \text{for all } t > 0. \tag{2}$$

Interpretation: The regulator treats offsets as units of a scalar wetland value. The empirical question is whether this scalar value misses important local dimensions such as flood protection.

3.2 Developer demand for offsets

A landowner in watershed h obtains a private value from developing wetlands:

$$u(X_{ht}, \xi_{ht}; \theta) + \epsilon_{it1} = \theta' X_{ht} + \xi_{ht} + \epsilon_{it1}. \tag{3}$$

With offset requirements and price P_t, the relative value of development is

$$u(X_{ht}, \xi_{ht}; \theta) - \tilde{\upsilon}_h \theta_P P_t + \epsilon_{it1} - \epsilon_{it0}. \tag{4}$$

Given Type-I extreme value shocks, aggregate offset demand is logit:

$$Q_t(P_t, W_t, X_t, \xi_t, \tilde{\upsilon}; \theta) = \sum_h \tilde{\upsilon}_h W_{ht} \frac{\exp(\theta' X_{ht} - \tilde{\upsilon}_h \theta_P P_t + \xi_{ht})}{1 + \exp(\theta' X_{ht} - \tilde{\upsilon}_h \theta_P P_t + \xi_{ht})}. \tag{5}$$

Interpretation: Offset prices enter development as a regulatory cost. Demand for offsets is therefore derived from landowners' demand to develop wetlands.

3.3 Demand estimation

The watershed-period demand estimating equation is

$$\ln \omega_{ht} - \ln(1 - \omega_{ht}) = \theta' X_{ht} + \theta_P \tilde{\upsilon}_h P_{ht} + \xi_{ht}, \tag{6}$$

where $\omega_{ht} = Q_{ht}/W_{ht}$ is the share of private wetlands developed.

- Price is instrumented using supply-side cost shifters.
- Instruments include historical sunk bank capacity, Hausman-style nearby-market cost shifters, and government conservation land purchases.
- The preferred estimate implies an average price elasticity close to -1.

Interpretation: The IV strategy uses variation in the cost of producing offsets to identify demand for wetland development.

3.4 Supply, inventory, and bank entry

Wetland banks enter, produce credits slowly, hold reserves, and sell credits over time. Let b_{ft} be credits released to bank f in period t, q_{ft} be credits sold, B_{ft} be reserves, and T_{ft} be bank age. The empirical trading policy is written as

$$q_{ft} = \chi(\hat{s}_t, B_{ft}, T_{ft}, \tilde{\upsilon}_f). \tag{7}$$

The paper disciplines predicted trades with static Cournot individual-rationality constraints:

$$1 + \frac{q_f^{IR}}{q_f^{IR} + \sum q_{-ft}} \cdot \eta(q_f^{IR} + \sum q_{-ft}, \hat{s}_t) = 0. \tag{8}$$

Entry costs are recovered by simulating value functions:

$$V(B_{f0}, T_{f0}, s_0, \tilde{\upsilon}_f) = \sum_{t=0}^{T} \delta^t \int_{\mathcal{S}^t} \Pi(\mathcal{Q}(s_t, B_{ft}, T_{ft}, \tilde{\upsilon}_f), s_t) dH^t(s_t | s_0). \tag{9}$$

Interpretation: Banks are dynamic oligopolists with inventories and delayed production. Observed entry and trading behavior reveal the fixed cost distribution needed to rationalize bank entry.

3.5 Flood damage specification

The flood externality model estimates expected claims using a quasi-Poisson specification:

$$\mathbb{E}[claims_{h,post}] = \exp(\zeta_d \ln Q_{h,1996-2016} + \zeta_b \, asinh(B_{h,1996-2016}) \tag{10}$$
$$+ \phi(claims_{h,pre}) + \gamma' X_h). \tag{11}$$

- Q_h is wetland development in watershed h.
- B_h is wetland bank area/restoration.
- ζ_d measures flood damage from wetland development.
- ζ_b measures flood protection from bank restoration.

Interpretation: The specification estimates local flood damages from wetland loss and local flood protection from restoration, after controlling for baseline flood risk and development.

3.6 Counterfactual market design

The main counterfactual adds a local Pigouvian correction equal to expected marginal flood damages:

$$Tax_h \approx EMD_h. \tag{12}$$

The paper compares:

1. observed market outcomes;
2. local Pigouvian taxes;
3. a uniform tax;
4. counterfactuals with alternative assumptions about passthrough and market structure.

Interpretation: The central design lesson is that the tax should vary across locations because flood externalities vary dramatically across watersheds.

4 Identification and empirical design

4.1 Demand identification

- Prices are endogenous because banks choose supply based on demand conditions.
- The instruments shift the cost or availability of offset supply while plausibly excluding local wetland development demand.
- Historical sunk capacity is especially strong because bank production capacity is fixed at entry and releases slowly.

Interpretation: The design identifies how much wetland development responds to offset prices, which is essential for evaluating taxes and counterfactual designs.

4.2 Supply-cost identification

- The paper observes who enters, where banks locate, and how much they produce over time.
- Entry decisions imply that entrants' value functions exceed entry costs, while nonentrants' costs exceed values.
- Forward simulation recovers value functions and conditional entry cost distributions.

Interpretation: Because observed banks are selected entrants, contract costs alone cannot identify the full cost distribution. The dynamic entry model corrects for selection.

4.3 Flood externality identification

- The paper compares watersheds with different wetland development and restoration histories.
- It controls for baseline claims, flood zones, baseline development, nonwetland development, demographics, and water-management fixed effects.
- The outcome is post-market flood claims for preexisting structures, reducing mechanical effects from new construction.

Interpretation: The flood estimates aim to identify the local damage caused by losing wetland flood protection rather than the flood risk of newly built structures.

4.4 Limits

- Counterfactual equilibria are approximated rather than solved in the full dynamic oligopoly model.
- Flood insurance claims understate total flood damages because many properties are uninsured or underinsured.
- The analysis emphasizes flood protection; other wetland services such as biodiversity, water quality, and carbon are not fully priced.
- Regulator wetland values may capture some ecological dimensions but not all location-specific services.

Interpretation: The paper's estimates are most persuasive for the trade-off between private gains from offset trade and local flood externalities, not for every possible ecological dimension.

5 Tables and figures

The figures and tables below are directly cropped from the paper and grouped to keep the notes compact.

5.1 Figure 1 and Table 1: geography and descriptive evidence

Read:

- Figure 1 shows Florida offset market boundaries, an example market, and spatial reallocation within that market.
- Wetland development occurs near existing development; wetland banks locate farther out, near existing wetlands.
- Table 1 documents large trade flows, concentrated markets, slow production, and flood-risk differences between development and restoration sites.
- Average annual prices are about $88,000 per offset, while observed restoration and land costs are much lower on average.

Economic message: Offset markets reallocate wetland conservation from dense development areas toward peripheral restoration sites. This creates private gains but can move flood protection away from people and property.

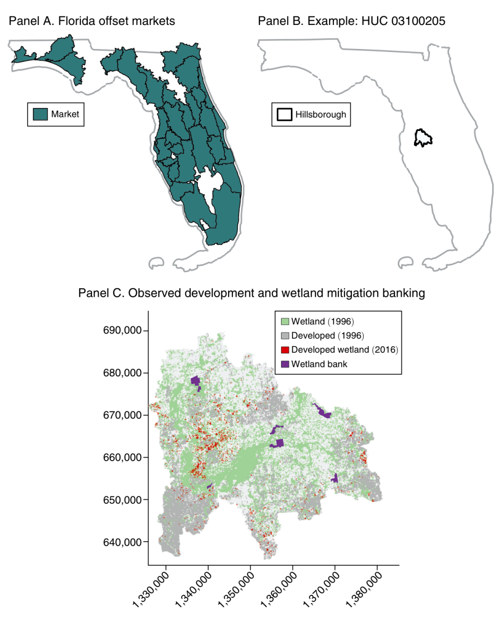


FIGURE 1. LOCATIONS OF WETLAND DEVELOPMENT AND RESTORATION

Notes: An example of our data on land use and wetland offsets within an offsets market. Initial wetland (green) and

Figure 1: locations of wetland development and restoration.

TABLE 1—NEW DATA ON WETLAND OFFSETS IN FLORIDA

	Observations	Avg	SD	q25	q50	q75
Initial wetlands[a] (pct/pixels)	136,302,645	36.5	48.1	0	0	100
Initial wetlands (pct/watershed)	1,004	34.0	20.1	20.0	30.8	44.0
Initial public land (pct/watershed)	1,004	12.5	21.8	0	2.1	14.7
Wetlands development and restoration						
Wetlands developed, 1996–2016 (acres)	1,004	207.5	483.4	2.4	16.3	186.7
Private wetlands developed	1,002	206.2	481.3	2.0	15.9	186.1
$\mathbb{P}$(develop|wet) × 100[b]	1,000	3.7	7.3	0.04	0.3	3.9
With wetland bank[c]	96	0.8	1.1	0.1	0.2	1.0
With high development[d]	179	15.1	10.1	7.0	14.2	20.8
Initial wetlands ('000 acres)	1,004	11.0	30.6	4.1	7.2	11.6
With wetland bank	96	23.1	76.1	7.6	10.5	15.9
With high development	179	10.1	13.8	3.5	7.0	11.2
Initial public wetlands ('000 acres)	1,004	3.3	25.0	0	0.3	1.9
With wetland bank	96	13.7	74.1	0	0.6	3.9
With high development	179	2.0	6.3	0.1	0.4	1.6
Initial private wetlands ('000 acres)	1,004	7.7	13.8	3.2	5.7	9.3
With wetland bank	96	9.4	5.5	5.8	8.4	11.2
With high development	179	8.1	8.8	3.3	6.0	9.6
Initial developed land (pct)	1,004	13.8	19.1	1.8	4.7	18.2
With wetland bank	96	5.7	6.6	1.2	3.0	8.0
With high development	179	37.6	23.0	19.1	32.5	55.7
Offset credit production and sales						
Bank entry year	107	2008.1	7.5	2003	2009	2014.5
Bank size (acres/bank)	107	1,866.1	2,680.0	428.5	1,049	2,157.5
Bank size (credits/bank)	106	410.0	566.1	85.2	203	521.8
1(credits released) per bank per year	1,209	0.3	0.4	0	0	1
Credits released (pct/bank/year)	343	15.3	16.2	5	10.0	20.0
Acres per credit	106	5.9	4.5	3.1	5.1	6.9
Acre wetland developed per credit sold	5,512	8.8	2.8	8.1	8.2	11.6
Annual sales (credits/bank-year)	981	15.5	31.4	0	1.8	15.4
Bank reserves (pct/bank-year)	967	51.8	33.6	18.3	54.7	82.0
Market structure						
Area ('000 acres/market)	30	1,153.3	800.4	516.8	863.4	1,520.4
Area (watersheds/market)	30	33.5	18.0	18.2	27.5	49
Entry (market-year)	780	11.7	32.1	0	0	0
Number of banks (market-year)	530	2.6	2.3	1	2	3
Offset price ('000$/credit), all transactions	1,432	87.5	61.7	38.6	63.4	137.2
Offset price ('000$/credit/market/year)	151	98.8	50.5	62.0	93.9	127.2
Flood risks						
Flood zone (pct/watershed)	1,004	41.7	23.8	23.9	37.3	56.1
Zone V (storm surge) (pct)	1,004	2.4	9.8	0	0	0
Zone A (100-yr) (pct)	1,004	39.4	22.5	23.0	35.6	51.7
Flood insurance claims[e] ('000$/claim)	188,368	31.3	71.1	3.3	10.5	32.9
Flood claims, pre-1996 ('000$/yr/watershed)	1,004	219.9	1,387.4	0	0.2	10.1
With wetland bank	96	314.6	2,345.2	0	0.1	4.7
With high development	179	412.7	1,552.4	0.8	10.7	99.4
Flood claims, post-2015 ('000$/yr/watershed)	1,004	335.2	1,414.6	0.003	4.9	71.5
With wetland bank	96	161.5	541.2	0.7	7.1	49.6
With high development	179	798.2	2,445.0	24.1	97.3	361.1

Notes: Descriptive statistics for Florida, 1995–2020. Supplemental Appendix Tables A2 and A3 contain additional data.

[a] Initial measures correspond to 1996 values.

[b] $\mathbb{P}$(develop|wet) defined as the within-pixel probability that a wetland pixel in 1996 becomes developed in

Table 1: descriptive statistics on Florida wetland offsets.

5.2 Tables 2–4: demand, supply costs, and flood damages

Read:

- Table 2 estimates demand for development on wetlands. IV estimates imply an average price elasticity close to -1 in the preferred specification.
- Table 3 estimates wetland bank entry costs. Average realized entry costs are about \$7 million per bank and about \$30,000 per credit, with large dispersion.
- Table 4 estimates local flood damage functions. Development on wetlands raises claims, while wetland bank area is associated with flood protection.
- Implied marginal damages per acre are highly skewed, with very large damages in the upper tail.

Economic message: The structural estimates supply the three objects needed for welfare analysis: demand for development, cost of restoration, and location-specific flood externalities.

Table 2—Estimated Demand for Development on Wetlands

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Credit price coefficient[a] (θ_P)	−0.34	−1.29	−0.98	−1.10	−1.45	−2.32	−1.06
	(0.14)	(0.28)	(0.26)	(0.38)	(0.60)	(0.58)	(0.39)
Implied parameters							
Average price elasticity	−0.3	−1.13	−0.85	−0.96	−1.31	−2.03	−0.96
Std dev price elasticity	0.15	0.58	0.44	0.49	0.66	1.03	0.48
Aggregate consumer surplus (bn US\$)	4.11	1.12	1.67	2.62	2.34	2.37	2.64
Instruments							
Historical sunk capacity		✓	✓	✓			✓
Hausman cost shifters					✓		✓
Government conservation land purchases						✓	✓
Additional controls							
Lagged demographics[d]			✓	✓	✓	✓	✓
HUC8 fixed effects[e]				✓	✓	✓	✓
First-stage F-stat		115.8	117.3	49.8	8.3	21.3	14.3
Observations	758	758	758	758	629	758	629
Adjusted R^2	0.70	0.68	0.70	0.71	0.68	0.64	0.70

Notes: Instrumental variable estimates of (14) at the watershed-by-period level for watershed-periods with prices and nonzero development. See Section IIIA for details. All columns include period and water district fixed effects and controls for baseline flood risk[b] and lagged development density.[c] Watersheds correspond to HUC12 units. Periods are (1996–2001, 2001–2006, 2006–2011, 2011–2016).

[a]Price coefficient from (14). Scaled by 1/100,000.
[b]Percent areas designated storm surge or 100-year flood zones.
[c]Share developed and share highly developed.
[d]Population and median income.
[e]Hydrological unit code (USGS 2013). All omitted coefficients reported in Supplemental Appendix Table A5.

Table 2: estimated demand for development on wetlands

Table 3—Estimated Wetland Bank Costs

	Observations	Mean	SD	q25	q50	q75
First-stage entry probabilities, $p_{\{enter\}}$	106	0.12	0.07	0.08	0.13	0.16
Value functions, $\mathbb{E}[V]$	106	17.88	29.99	1.37	4.82	15.91
Parameter estimates						
$\mu_\kappa(s_{mt})$	106	15.98	2.56	14.46	15.12	16.12
$\sigma_\kappa(s_{mt})$	106	1.17	1.68	0.12	0.94	1.19
Implied costs						
Realized entry cost estimate (MM/bank)	106	7.01	10.71	1.06	2.85	6.85
Est. entry costs per credit ('000/bank)	106	29.93	41.97	5.79	13.66	35.53
Implied rate of return on capital (pct)	106	6.08	5.97	1.79	3.81	8.30
Comparison with contract data						
Observed entry costs (MM/bank)	79	5.29	6.09	1.42	2.86	7.18
Observed entry costs per credit ('000/bank/credit)	79	23.95	23.27	9.20	15.99	31.17
Observed construction costs (MM/bank)	86	1.61	2.50	0.36	0.97	1.81
Observed land costs (MM/bank)	95	5.05	10.53	0.57	1.89	5.53

Notes: Wetland bank cost estimates. See Section IIIB for details. Additional results appear in Supplemental Appendix Table A8.

Table 3: estimated wetland bank costs

TABLE 4—ESTIMATED LOCAL FLOOD DAMAGE FUNCTIONS

	(1)	(2)	(3)	(4)
Development on wetlands (ζ_d)	0.428	0.245	0.243	0.147
	(0.077)	(0.083)	(0.084)	(0.085)
Wetland bank area (ζ_b)	−0.083	−0.093	−0.093	−0.108
	(0.042)	(0.036)	(0.036)	(0.034)
Flood Zone V (storm surge) (%)		2.922	2.919	1.776
		(0.892)	(0.895)	(0.949)
Flood Zone A (100-yr) (%)		0.849	0.859	1.390
		(0.824)	(0.831)	(0.583)
Nonzero baseline flood claims (1991–1995)		3.069	2.700	2.407
		(0.417)	(0.395)	(0.288)
Baseline flood claims (1991–1995)		0.236	0.236	0.081
		(0.097)	(0.097)	(0.117)
Baseline flood claims (1991–1995) squared		−0.009	−0.009	0.001
		(0.007)	(0.007)	(0.008)
Additional controls				
Demographic controls		✓	✓	✓
HUC8 FEs				✓
Implied damages ($/acre)				
0%	135.7	6.7	9.7	0.000
10%	5,625.1	873.6	1,065.1	124.2
25%	16,624.0	2,977.9	3,533.3	589.1
50%	54,315.2	10,145.9	12,746.6	2,614.4
75%	184,120.7	39,954.6	52,343.7	11,588.8
90%	391,284.0	151,036.6	184,177.3	38,930.7
95%	574,790.8	275,008.5	369,786.0	89,572.5
97.5%	744,815.1	501,671.4	576,853.7	195,402.1
99%	797,514.3	754,819.5	855,061.2	336,950.8
99.9%	1,076,979.0	4,792,847.0	5,256,975.0	1,531,050.0
100%	10,579,423.0	5,380,962.0	5,369,403.0	2,257,844.0
Observations	1,226	1,226	1,015	1,226

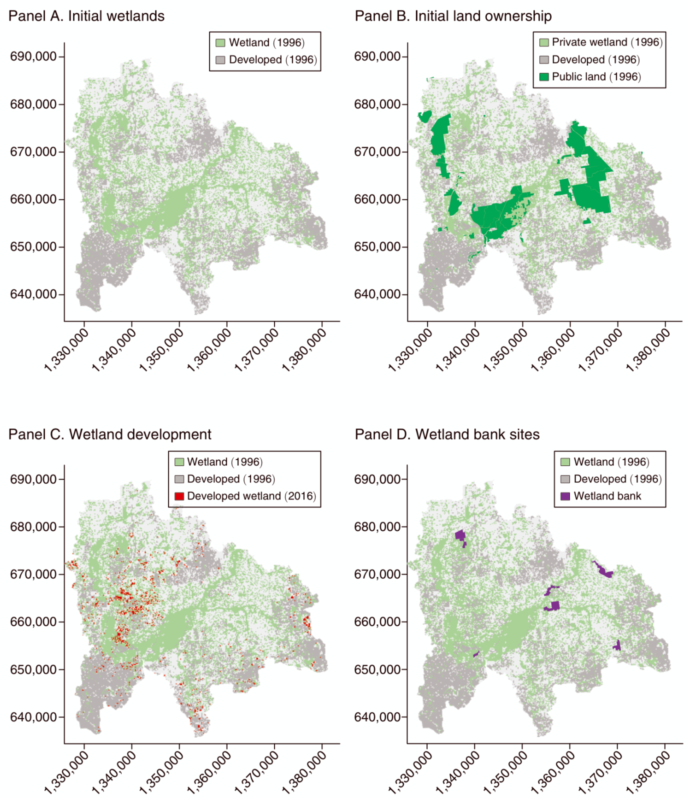
Notes: Quasi-Poisson estimates of (20) at the local watershed level for watersheds with nonzero wetland development. All columns include water district fixed effects and controls for baseline development density and other development on nonwetlands. The outcome is flood insurance claims after the market (2016–2020) for properties built prior to the market (1995); see Supplemental Appendix Table A12 for other outcomes. Column 3 restricts the sample to watersheds with nonzero flood insurance policies in 1995. Implied damages report quantiles of watershed-level expected marginal damages (at observed development) per acre wetland developed. All omitted coefficients reported in Supplemental Appendix Table A9. Robust (HC1) standard errors clustered at the HUC12 level in parentheses.

5.3 Figures 2–4: baseline geography, development values, and bank costs

Read:

- Figure 2 decomposes the same example market into initial wetlands, land ownership, wetland development, and wetland bank sites.
- Figure 3 maps observed wetland development and plots estimated developer values by trade.
- Figure 4 maps bank entry probabilities and plots estimated producer surplus and costs by offset credit.
- Developer values and bank costs are both heterogeneous, creating private gains from trade.

Economic message: These figures show why markets can be valuable: development pressure and restoration opportunities are spatially different.



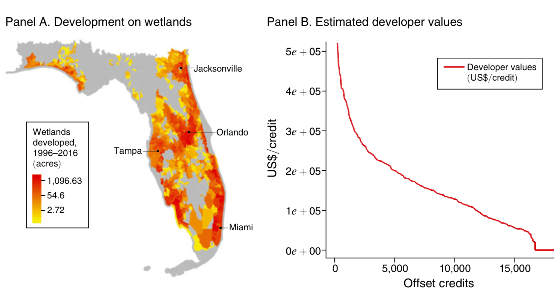


FIGURE 3. DEVELOPMENT ON FLORIDA WETLANDS

Notes: Panel A: Map of local watershed development occurring in offset markets between 1996 and 2016. Local watersheds colored by decile of ln(acres of wetlands developed). Panel B: Estimated private values for land developers who purchased offsets, calculated with (21), ordered left to right by trades' decreasing estimated value, 1995–2018.

Figure 3: development on Florida wetlands.

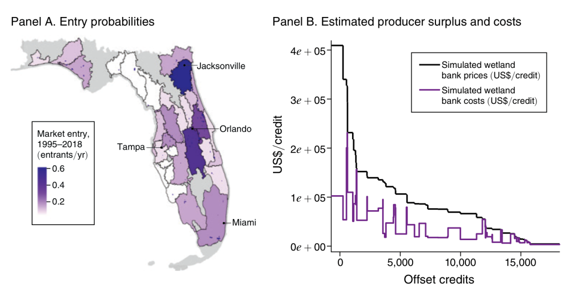


FIGURE 4. WETLAND MITIGATION BANKS

Notes: Panel A: Map of average annual market entry probabilities and wetland bank sites. See Supplemental Appendix Figure A4 for variation across market-years. Panel B: Estimated per credit costs and transaction values for wetland banks, calculated with equations (A2) and (A3) and ordered left to right by increasing simulated price per credit.

Figure 4: wetland mitigation banks.

5.4 Table 5 and Figure 5: welfare and flood externalities

Read:

- Table 5 reports the welfare decomposition for the observed market, a local Pigouvian tax, and a uniform tax.
- The observed market produces private gains from trade of about \$2.41 billion but also about \$1.89 billion in flood damages.
- Net welfare under the observed market is about \$522 million.
- The local Pigouvian design raises welfare to about \$1.78 billion by sharply reducing flood damages while retaining much of the gains from trade.
- Figure 5 shows that flood damages are highly spatially skewed and concentrated in a subset of wetland-

development trades.

Economic message: The market's private gains are large, but so are unpriced external damages. Because damages are local and skewed, a uniform correction performs worse than a local Pigouvian policy.

TABLE 5—WELFARE AND OFFSET MARKET DESIGN

	Market	Pigou	Uniform tax
Wetlands developed (acres)	141,606.2	120,097.6	75,168.8
Wetlands offsets used (credits)	16,694.3	14,256.0	8,922.2
Gains from trade			
Developer values (MM)	2,850.6	2,486.2	2,193.0
Supply costs (MM)	440.3	421.4	361.6
Private gains from trade (MM)	2,410.3	2,064.9	1,831.3
Distributional outcomes			
Consumer surplus (MM)	1,678.1	1,339.3	841.6
Producer surplus (MM)	732.2	540.7	124.2
Tax revenue (MM)	0	184.8	865.5
Externalities			
Flood damage (MM)	−1,888.1	−282.1	−714.6
Below 99.9 percentile	−1,888.1	−282.1	
Below 99 percentile	−1,719.5	−282.1	
Below 97.5 percentile	−1,702.4	−284.9	
7% discount rate	−1,132.9		
3% discount rate	−2,643.4		
Welfare (MM)	522.2	1,782.8	1,116.7

Notes: Values in millions of 2020 US dollars. Market outcomes from 1995 to 2020 at the observed equilibrium (column 1), counterfactual equilibrium with local Pigouvian taxes (column 2), and counterfactual equilibrium with the uniform tax that maximizes the sum of private gains from trade and total flood benefits from conservation (column 3). Net present discount values calculated using a 5 percent real discount rate. The uniform tax is calculated to maximize the difference between net surplus and insured flood damages; its optimal level is calculated to be $97,000/offset.

Table 5: welfare and offset market design.

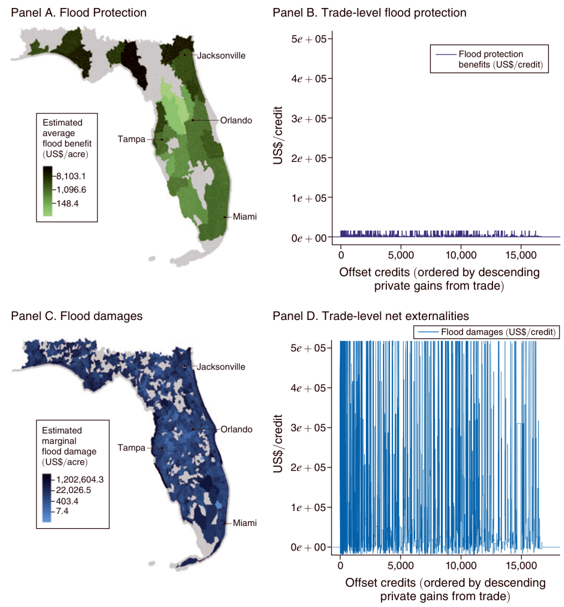


FIGURE 5. REALIZED FLOOD EXTERNALITIES

Notes: Panel A: Map of estimated market-level flood protection benefits from wetland banks. Panel B: Estimated average flood protection benefit for each wetland under the market from 1996 to 2016, calculated with (19) and sorted by descending private gains from trade. Panel C: Map of estimated marginal flood damages at the watershed level for

Figure 5: realized flood externalities.

5.5 Figure 6 and Table 6: Pigouvian redesign and market structure

Read:

- Figure 6 overlays flood damages and private gains under observed trades, a local Pigouvian tax, and a uniform tax.
- The local Pigouvian tax screens out high-damage trades while preserving many high-private-gain trades.
- Table 6 shows robustness to alternative assumptions about passthrough and market structure.
- Pigouvian policies reduce flood damages and improve welfare under benchmark, myopic Cournot, and myopic collusion assumptions.

Economic message: The policy message is robust: even when market conduct and passthrough differ, pricing local flood damages improves welfare while preserving substantial gains from trade.

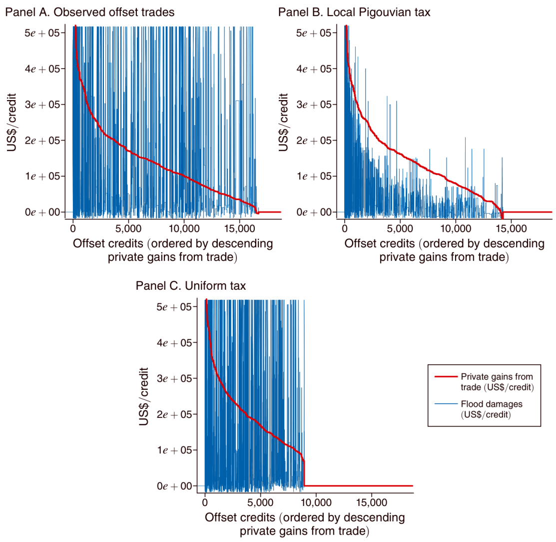


FIGURE 6. PIGOUVIAN REDESIGN

Notes: Panel A: Estimated flood damages from Figure 5, panel D, plotted against the private gains from trade (i.e., the difference between the developer values and bank costs in Supplemental Appendix Figure A7). Panel B: Estimated private gains from trade and flood damages under the Pigouvian flood protection taxes at the local watershed level, sorted by descending private gains from trade. Panel C: Estimated private gains from trade and flood damages under a uniform tax that maximizes the sum of private gains from trade net of total flood damage, sorted by descending private gains from trade. See Section IVC for details.

Figure 6: Pigouvian redesign.

TABLE 6—WELFARE, PASSTHROUGH, AND OFFSET MARKET STRUCTURE

	Benchmark		Myopic cournot		Myopic collusion	
	Market	Pigou	Market	Pigou	Market	Pigou
Wetlands developed ('000 acres)	141.6	120.1	209.5	176.0	140.6	114.2
Wetlands offsets ('000 credits)	16.7	14.3	24.9	21.2	16.4	13.5
Total transaction volume (MM)	1,172.5	962.1	1,468.6	1,167.7	1,831.9	1,509.0
Passthrough						
Average price ('000$/credit)	70.2	67.5	58.9	55.2	111.4	111.4
Average price + tax ('000$/credit)	70.2	80.5	58.9	67.6	111.4	122.8
Producer price change (%)		−3.9		−6.2		−0.02
Consumer passthrough (%)		78.8		70.3		99.8
Gains from trade						
Developer values (MM)	2,850.6	2,486.2	3,969.2	3,395.6	3,470.5	2,885.3
Supply costs (MM)	440.3	421.4	485.1	473.2	425.8	415.8
Private gains from trade (MM)	2,410.3	2,064.9	3,484.1	2,922.4	3,044.7	2,469.5
Distributional outcomes						
Consumer surplus (MM)	1,678.1	1,339.3	2,500.6	1,966.2	1,638.5	1,221.3
Producer surplus (MM)	732.2	540.7	983.5	694.6	1,406.1	1,093.2
Producer surplus (%GFT)	30.4	26.2	28.2	23.8	46.2	44.3
Tax revenue (MM)	0	184.8	0	261.7	0	155.1
Producer surplus change (%)		−26.2		−29.4		−22.3
With lump-sum transfer (%)		−0.9		−2.8		−11.2
Externalities						
Flood damage (MM)	−1,888.1	−282.1	−2,752.0	−554.7	−1,798.0	−253.4
Damages (% pre-reform)		14.9		20.2		14.1
Change (% welfare change)		−465.0		−391.2		−268.6
Change (% GFT change)		127.4		134.3		159.3
Welfare (MM)	522.2	1,782.8	732.1	2,367.7	1,246.7	2,216.1

Notes: Market outcomes from 1995 to 2020 at (i) benchmark offset trading policy functions, (ii) benchmark trading policy functions with local Pigouvian taxes, (iii) myopic Cournot trading policy functions, (iv) myopic Cournot trading policy functions with local Pigouvian taxes, (v) myopic collusion trading policy functions, and (vi) myopic collusion trading policy functions with local Pigouvian taxes. Consumer passthrough (%) is defined as [the average posttax producer price, plus the tax, minus the average pretax producer price] divided by the average tax. Producer surplus change (%) [with lump-sum transfer] defined as the percentage change in producer surplus [plus total tax revenue] relative to previous column. Flooding (%, pre-reform) reports the flooding as a percent of flooding in the previous column; changes (%) report counterfactual changes in flood damage relative to the previous column as percentages of the changes in welfare and private gains from trade.

Table 6: welfare, passthrough, and market structure.

6 Short summary

- The paper develops an empirical framework for evaluating environmental offset markets.
- The application is Florida wetland mitigation banking from 1995 to 2020.
- Offsets allow developers to destroy wetlands if they purchase credits from wetland banks that restore wetlands elsewhere.
- The market generates large private gains because development and restoration opportunities differ spatially.
- The market also creates flood externalities because development tends to occur near exposed property, while restoration occurs in peripheral areas.
- Demand estimates imply meaningful responsiveness of wetland development to offset prices, with price elasticity near -1.
- Supply estimates recover wetland bank entry costs and dynamic trading behavior.
- Flood estimates show that local flood damages from wetland development are highly skewed.
- The observed market generates about \$2.4 billion in private gains but about \$1.9 billion in flood damages.
- A local Pigouvian tax prevents much of the flood damage while preserving more than two-thirds of private gains from trade.
- The broad lesson is that offset markets require spatially differentiated prices when the public good has local services.

7 Conclusion

7.1 Core conclusion

Aronoff and Rafey show that environmental offset markets can generate large private gains from trade, but these gains do not imply social efficiency. In Florida wetland markets, trading reallocates conservation away from places where wetlands provide valuable local flood protection. This creates large flood externalities that are not priced under the existing offset design.

7.2 Economic intuition

Offset markets work well for the regulator's scalar No Net Loss goal when one unit of restored wetland value is a good substitute for one unit of lost wetland value. The problem is that wetlands also provide local services. A wetland near housing may protect valuable property from floods, while a restored wetland in a peripheral location may not provide the same local protection. The private offset price does not reflect this spatial difference, so development overuses high-externality wetlands.

7.3 Takeaway

The paper's main design lesson is that environmental offsets should not be treated as undifferentiated permits when environmental services vary across locations. A locally differentiated Pigouvian tax can preserve most private gains from trade while preventing a large share of environmental damage.